Mohammed Rupawala

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Education

Adani University — **Ahmedabad, Gujarat, India**
B.Tech in Information and Communication Technology — *Aug 2022 – Present*

- CGPA: **7/10**
- Coursework: Data Structures and Algorithms, Operating Systems, Database Management Systems, Computer Networks, Machine Learning, Deep Learning, Computer Vision

Technical Skills

Languages: JavaScript, TypeScript, Python, C++, SQL, GO, HTML/CSS
Technologies: Node JS,FastAPI, React JS, NextJS, Tailwind
Tools: Vs Code, Docker, Git, GitHub

Experience

E2M Solutions — **Jan 2026 – Present**
AI Engineering Intern

- Designed and launched a multi-tenant SaaS AI chatbot platform used by 20+ client organizations, enabling plug-and-play knowledge base ingestion (PDFs, docs, URLs).
- Built end-to-end ingestion and retrieval pipelines using embeddings and semantic search, reducing client support workload by 80–90% and improving response accuracy across domains.

Projects

CUTI | *GO,Linux* — **Feb 2026**

- Built a high-performance TCP server in Go that can handle 10,000+ users at the same time efficiently, using an event-driven architecture similar to Redis.
- Implemented Redis's communication protocol (RESP) from scratch, enabling faster command processing and improving overall request speed by 3–4x.
- Engineered a crash-safe data persistence system utilizing an Append-Only File design, achieving 99.9% data integrity during server outages.

Deki | *Go, OS, CLI* — **Jan 2026**

- Built a Unix-like shell in Go with support for piping, I/O redirection, and 10+ built-in commands.
- Integrated external command execution (e.g. git, ls) using os/exec for full CLI functionality.
- Implemented Trie-based autocomplete for efficient prefix matching and command suggestions.

MindSync | *React, Tailwind, FastAPI, TypeScript, PostgreSQL, Redis, LangChain* — **Apr 2025**

- Built an AI-powered journaling platform using RAG and vector embeddings, enabling context aware search and conversational querying across 100+ journal entries with 6–7s retrieval latency.
- Integrated a Hugging Face–based sentiment analysis model to perform multi-class mood classification on journal entries of up to 1,000+ words, extracting actionable emotional insights.
- Optimized the retrieval and inference pipeline to parse, analyze, and respond to user queries within 7 seconds end-to-end, balancing accuracy and latency for real-world usage.

Hackathons

Hackspire: Finalist (Top 5), Adani University — Apr 2025

- Built a smart queue management system using demand forecasting to predict optimal arrival times, reducing user wait times by 10+ minutes and improving service efficiency.

Odoo Hackathon: Final Round, GIFT City, Gujarat — Jul 2025

- Developed a Q&A platform with question tagging, threaded answers, and voting, enabling efficient knowledge sharing within organizations.

Competitive Programming

LeetCode(THFC): 1650 — Codeforces(Special_One): 1200